

西方经济学

Part 3 Growth Theory

THE ECONOMY IN THE VERY LONG RUN

Lecture 3 The Solow (1956)–Swan (1956) Model

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- (1) M16; S9.¹
- (2) 其他资料: Loyalka, P. et al. 2021. [Skill levels and gains in university STEM education](#) in China, India, Russia and the United States. *Nature Human Behaviour* 5, 892–904.
- (3) 其他资料: [GM]
 - CAPITAL ACCUMULATION AS A SOURCE OF GROWTH
 - POPULATION GROWTH AND TECHNOLOGICAL PROGRESS
 - GROWTH EMPIRICS AND POLICY



¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

学习目标

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- (1) 理解长期增长的动力。
- (2) 掌握稳态概念，分析稳态的改变。
- (3) 掌握条件收敛与收敛速度。
- (4) 了解增长核算的 Solow 残差法。
- (5) 领会马工程教材精神。



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Outline

1 Definition of Economic Growth

2 The Basic Solow Model

- The Fundamental Equation
- Actual Investment
- The Balanced Growth Path
- The Effect of Raising Saving Rate s
- The Golden Rule Level of Capital
- Transitional Dynamics
- Extension: Population Growth and Technological Progress

3 Growth Empirics

- Convergence
- Growth Accounting

4 马工程教材疑难重点

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Definition of Economic Growth

马工程观点 (M, 2019, p.249)

一、经济增长和经济发展

从本书第九章中可知，在宏观经济学中，国内生产总值（GDP）既是衡量一个国家（或地区）经济活动的重要指标，也是反映该国（或该地区）在一定时期内生产总成果的重要指标。因此，从理论的层面看，为了描述和反映一个经济体（国家或地区）物质产品的丰富和增加，很自然地联系到以 GDP 表示的产量的概念。

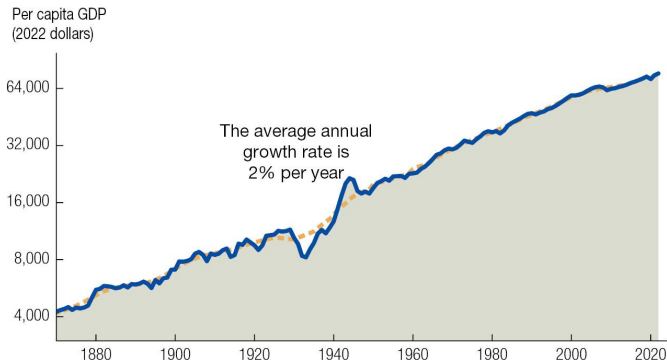
一般地，在宏观经济学中，经济增长被定义为产量的增加，这里，产量既可以表示为经济的总产量（GDP 总量），也可以表示为人均产量（人均 GDP）。经济增长的程度可以用增长率来描述。



Definition of Economic Growth

Jones's (2025) Figure 1.7

Per Capita GDP in the United States



The growth rate of a discrete-time variable x is defined as $\frac{x(t+\Delta t) - x(t)}{\Delta t \cdot x(t)}$. The growth rate of a continuous-time variable x is defined as $\lim_{\Delta t \rightarrow 0} \frac{x(t+\Delta t) - x(t)}{\Delta t \cdot x(t)}$.



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The Basic Solow Model

The Fundamental Equation

Assumption 1

The production function, $F(K, N)$, is neoclassical.

Assumption 2

Labor input N is constant. Technology index A is also constant.

Let $y = Y/N$, $k = K/N$.

$$Y = F(K, N) \Rightarrow y = f(k)$$

With the assumption of $S = I$, the fundamental equation is

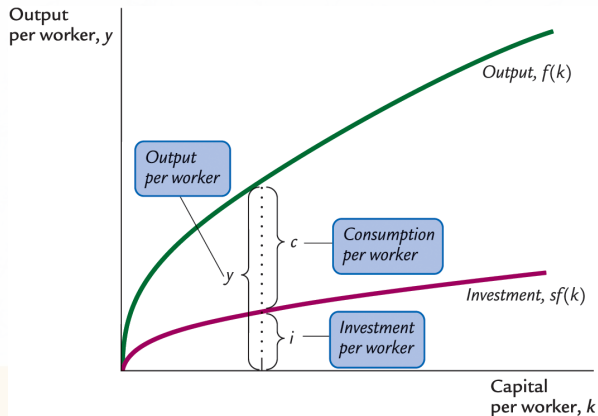
$\dot{K}(t)$ = Investment – Depreciation.

$$\dot{K}(t) = s \cdot Y(t) - \delta \cdot K(t).$$

$$\dot{k}(t) = s \cdot f(k(t)) - \delta \cdot k(t).$$



Actual Investment



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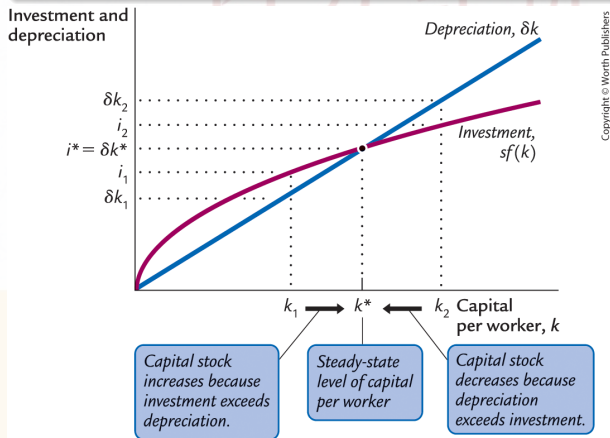


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The Balanced Growth Path

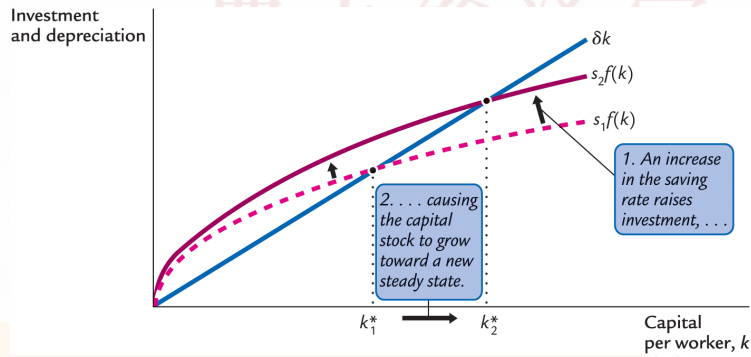
Definition 1

A *balanced growth path* is defined as a situation in which various quantities grow at constant rates. It is also called the *long run state* or *steady state*.



The Effect of Raising Saving Rate s

Is Capital Accumulation a Driving Force?



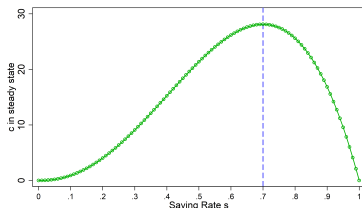
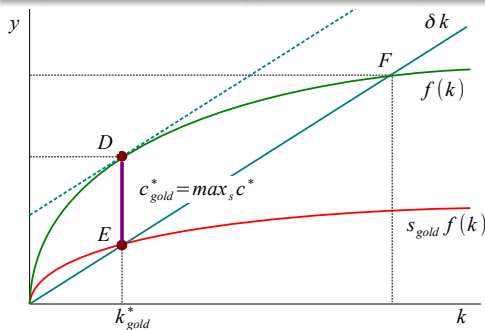
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The Golden Rule Level of Capital

Definition 2

The policymaker chooses the saving rate s such that the steady state value of consumption per worker ($c = C/N$) is maximized. The optimal saving rate is called the golden rule level of the saving rate, and is denoted by s_{gold} . The steady state value of capital per worker corresponding to s_{gold} is called the golden rule level of capital, and is denoted by k_{gold}^* .

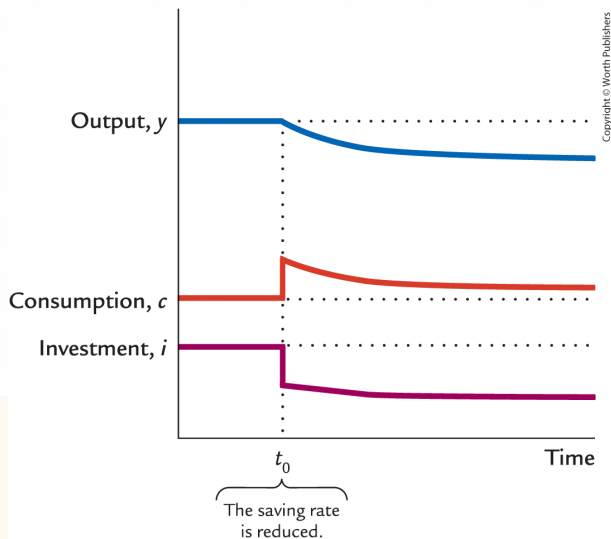


Note: depreciation rate = .1; production function is given by $\ln(y) = a \ln(k)$, where $a = .7$.

Golden Rule Level of Consumption per Capita

Transitional Dynamics

Starting with Too Much Capital ($k_0 > k_{Gold}^*$) and Lowering s



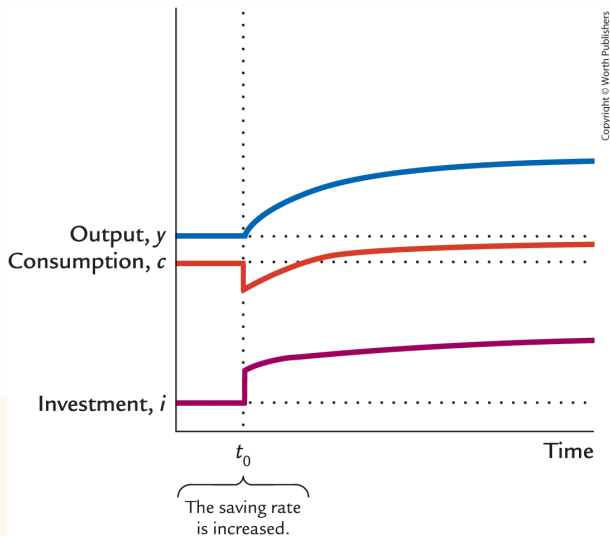
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Transitional Dynamics

Starting with Too Little Capital ($k_0 < k_{Gold}^*$) and Raising s



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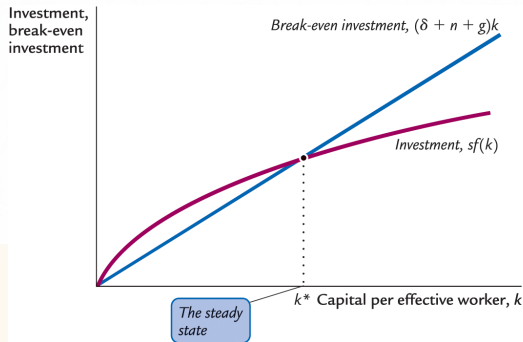


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Extension

Population Growth and Technological Progress

Suppose that $N(t) = N(0)e^{nt}$ and $A(t) = A(0)e^{gt}$. The production function is $Y = F(K, AN)$, where AN is called the effective worker.² The fundamental equation becomes $\dot{\hat{k}} = sf(\hat{k}) - (n + \delta + g)\hat{k}$, where $\hat{k} = K/(AN)$, $\hat{y} = f(\hat{k}) = Y/(AN)$.



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Example 1

What is the effect of raising n or g ?
What is the golden rule level of capital per effective worker?

² Effective: Actual. // Effective demand. (Merriam-Webster).

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Convergence

Conditional Convergence

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1. If poor economies will tend to catch up with rich economies, this process is called (*absolute*) *convergence*.
2. By the Solow model, we expect convergence among a group of economies which have the same steady state.
3. If a group of economies have different steady states, the Solow model predicts that they converge to their own steady states. This kind of convergence is called *conditional convergence*.



Convergence

The Speed of Convergence

Definition 3

Suppose that an economic variable $x(t)$ converges to a constant long-run equilibrium x^* . Let $\tilde{x}(t) = x(t) - x^*$. The *speed of convergence* of $x(t)$, defined as $\lim_{\Delta t \rightarrow 0} \left| \frac{\tilde{x}(t+\Delta t) - \tilde{x}(t)}{\Delta t \tilde{x}(t)} \right|$, measures how quickly $\tilde{x}(t)$ converges to 0 over time. The speed of convergence of a system is defined as the speed of convergence of a variable which has a constant long-run equilibrium in the system.

In the neighborhood of $\widehat{k}^*$,

$$\dot{\widehat{k}} = sf(\widehat{k}) - (n + g + \delta)\widehat{k} = -\lambda^* \cdot (\widehat{k} - \widehat{k}^*) + o(\widehat{k} - \widehat{k}^*),$$

where $\alpha_K^* = \frac{f'(\widehat{k}^*)\widehat{k}^*}{f(\widehat{k}^*)}$, and $\lambda^* = (1 - \alpha_K^*)(n + \delta + g)$ is the speed of convergence of the linearized system.

Barro (2015) finds that the speed of convergence based on international data is about 2 percent per year.



Suppose the production function is $Y = AF(K, N)$. In growth accounting, A is often called the *total factor productivity* (TFP) which captures effects on output level caused by factors other than K and N . The decomposition of the growth rate of output is given by

$$\frac{\dot{Y}}{Y} = \frac{\dot{A}}{A} + \frac{KF_K}{F} \frac{\dot{K}}{K} + \frac{NF_N}{F} \frac{\dot{N}}{N}. \quad (1)$$

Let $\alpha_K \triangleq \frac{KY_K}{Y}$, $\beta_N \triangleq \frac{NY_N}{Y}$ denote the factor income shares. Let χ_{TFP} denote the contribution of TFP.

$$\chi_{TFP} = g_Y - \alpha_K g_K - \beta_N g_N.$$

If F is homogeneous of degree one in K and N , then $\alpha_K + \beta_N = 1$.

Example 4

If $Y = F(K, AN)$, find the decomposition of the growth rate of Y similar to (1).



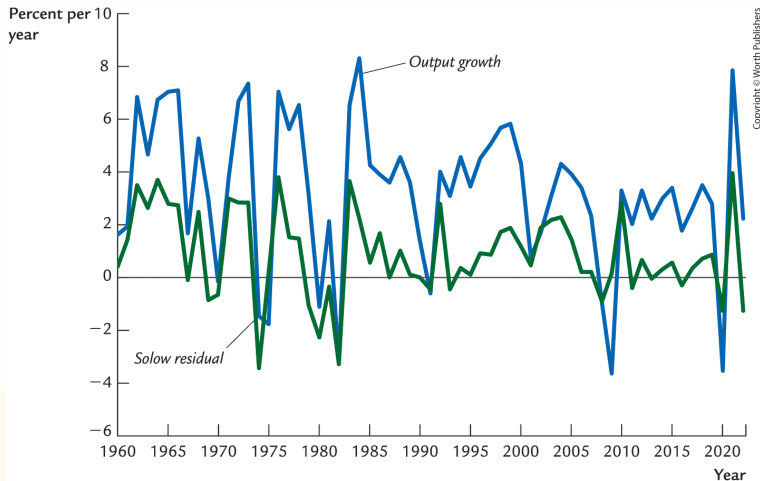
The Slowdown in Productivity Growth

TABLE 11-1 Accounting for Economic Growth in the United States

SOURCES OF GROWTH							
Years	Output Growth $\Delta Y/Y$	=	Capital $\alpha \Delta K/K$	+	Labor $(1-\alpha)\Delta L/L$	+	Total Factor Productivity $\Delta A/A$
(average percentage increase per year)							
1948–2022	3.4		1.3		1.0		1.1
1948–1973	4.2		1.3		1.0		1.9
1973–2022	2.9		1.3		1.0		0.6

Data from : U.S. Department of Labor. Data are for the nonfarm business sector. Parts may not add to total due to rounding.

The Solow Residual



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ARTICLES

<https://doi.org/10.1038/s41562-021-01062-3>

nature
human behaviour



Skill levels and gains in university STEM education in China, India, Russia and the United States

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Universities contribute to economic growth and national competitiveness by equipping students with higher-order thinking and academic skills. Despite large investments in university science, technology, engineering and mathematics (STEM) education, little is known about how the skills of STEM undergraduates compare across countries and by institutional selectivity. Here, we provide direct evidence on these issues by collecting and analysing longitudinal data on tens of thousands of computer science and electrical engineering students in China, India, Russia and the United States. We find stark differences in skill levels and gains among countries and by institutional selectivity. Compared with the United States, students in China, India and Russia do not gain critical thinking skills over four years. Furthermore, while students in India and Russia gain academic skills during the first two years, students in China do not. These gaps in skill levels and gains provide insights into the global competitiveness of STEM university students across nations and institutional types.

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- (1) 提高储蓄率对稳态人均收入有何影响？提高人口增长率对稳态人均收入有何影响？
- (2) 人均收入长期增长的动力来自哪里？
- (3) What is conditional convergence?
- (4) What is the difference between the rate of change and the growth rate of a variable?
- (5) 在 Solow 模型的稳态下，水平变量、人均变量和有效人均变量的增长率分别是什么？
- (6) Solow 模型的基本方程是什么？
- (7) 稳态的决定因素有哪些？每个因素如何影响稳态有效人均资本？
- (8) 稳态附近收敛速度的推导和直觉解释。
- (9) Growth accounting 的分解表达式。
- (10) What is the Solow residual?



马工程教材疑难重点

1 (E2, p.281)

根据马工程教材观点，应当如何评析西方经济学的经济增长理论？

2 (E2, p.284)

根据马工程教材观点，经济增长理论是否可以说明中国提出的创新驱动发展战略？

3 (E2, p.28)

根据马工程教材观点，应当如何看待西方经济学的研究方法？

